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Paleomagnetism of Permian Rocks of the Subpolar Urals, Kozhim River: to the History of Evolution of the Thrust Structures in the Subpolar Urals

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Abstract—The collections of Permian rocks from sections of the Kozhim River (Asselian, Kungurian, and Ufimian stages) and the Kama River (Ufimian and Kazanian stages) are studied. The paleomagnetic directions determined on the studied structures closely agree with the existing data for the Subpolar Urals and Russian Platform (RP). In the Middle Permian red clays of the Kama River region, the paleomagnetic pole $N/n = 28/51$, $\Phi = 47°$ N, $\Lambda = 168°$ E, $dp = 3°$, and $dm = 5°$ is obtained. The analysis of the existing paleomagnetic determinations for the Early and Middle Permian of the Russian and Siberian platforms and Kazakhstan blocks (KBs) is carried out. For the Subpolar Ural sections, the estimates are obtained for the local rotations during the collision of the Uralian structures with the Russian and Siberian platforms and KBs. The amplitudes of the horizontal displacements of the studied structures are, on average, 170 ± 15 km per Middle Permian. The scenario describing the evolution of the horizontal rotations of the structures of Subpolar Urals is suggested.

Keywords: thrusts, paleomagnetic belts, horizontal rotations, anisotropy of magnetic susceptibility, Subpolar Urals

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INTRODUCTION

The paleomagnetic investigations of the Subpolar Urals Paleozoic rocks that were recently carried out at the laboratory of magnetostratigraphy and paleomagnetic reconstructions of the All-Russia Petroleum Research Exploration Institute (VNIGRI) have shown that the pre-Permian Paleozoic rocks of this region have been remagnetized at the time of existence of the Kiama hyperchron (C_2–P_2) due to the magnetoviscous processes during the protrusion of these deposits to the surface under the collision of the East European Platform (EEP) with the Uralian structures (Iosifidi and Khramov, 2013). The closeness of the formation time of the orogenic dislocations to the time of remagnetization of pre-Late Paleozoic rocks in the Subpolar Urals allows the metachron components of natural remanent magnetization to be used for estimating the horizontal motions of the Paleozoic structures (Eremenko et al., 2009; Iosifidi and Khramoiv, 2013) and reconstructing ancient sedimentation basins in the Middle Permian. The horizontal rotation angles of the studied structures were estimated with the use of the reference paleomagnetic pole for the boundary between the Early–Middle Permian and the time of Late Paleozoic tectonic activation in the Urals (Kuznetsov et al., 2000). These estimates raise the requirements to the reliability on the reference paleomagnetic poles. Thus, a paradoxical situation has arisen: the paleomagnetic data obtained from the active tectonic regions are more reliable than those over the stable Russian Platform (RP).

The paleomagnetic investigations of the Permian rocks of the RP started at the end of the 1950s (Khramov, 1958; 1960). During the period of the paleomagnetic boom in the former Soviet Union in the 1960s–1970s, more than 100 determinations were obtained for the Permian rocks of the RP (*Paleomagnitnye...*, 1971; 1973; 1975; 1979; 1982; *Dannye...*, 1984; 1986; 1989). The reliability of these data does not always satisfy the present-day reliability criteria of paleomagnetic determinations. This is primarily associated with the lack of the required instruments and equipment at the Soviet paleomagnetic laboratories in the 1960s–1970s. The analysis of the existing data shows that most of the determinations were done on red rocks. On one hand, this requires thermal demagnetization of the natural remanent magnetization (NRM) up to the deblocking temperatures of hematite and the subsequent principal component analysis of the NRM composition. On the other hand, the recent studies on a number of the previously investigated

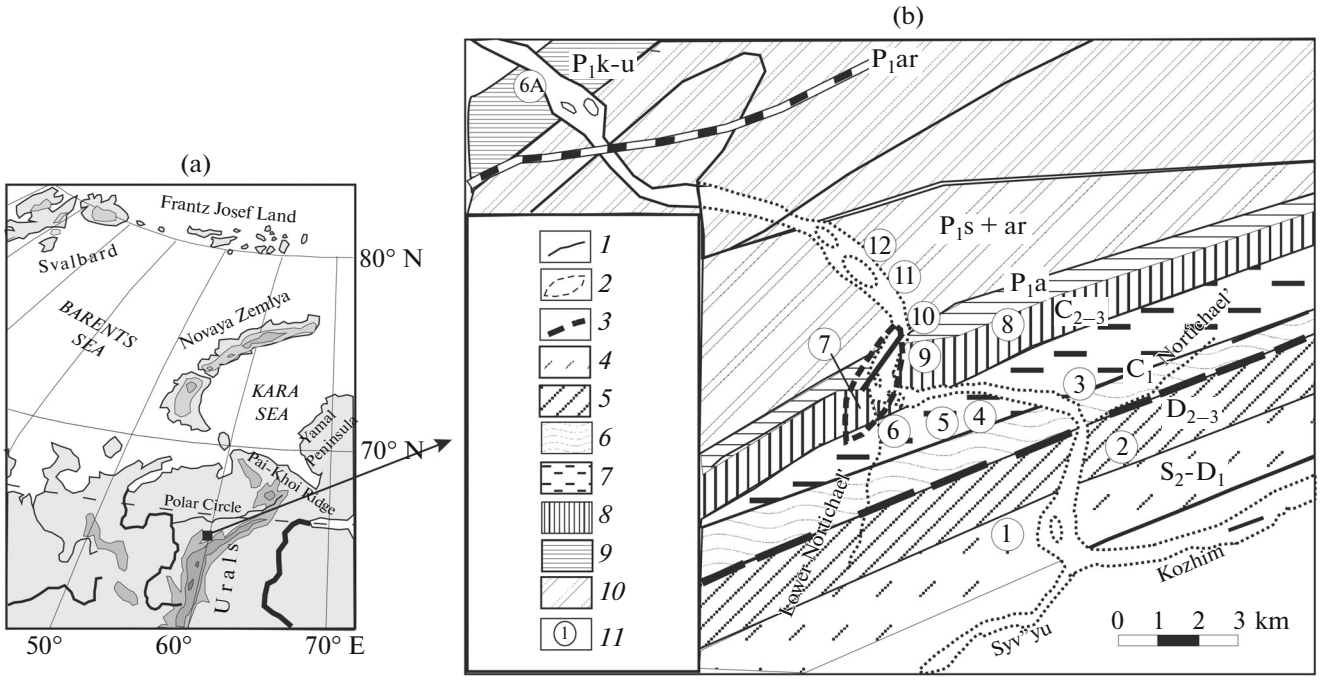


Fig. 1. Layout of studied sections and scheme of sampling: (a) region of study; (b) geological scheme of outcrops on Kozhim River: *1.* boundaries of formations; *2.* tectonic relic; *3.* tectonic boundaries; age of sediments: *4.* Late Silurian–Early Devonian; *5.* Middle- and Late Devonian; *6*, *7.* Early Carboniferous; *8.* Middle and Late Carboniferous; *9*, *10.* Early Permian; *11.* outcrop numbers (*Opornyi*..., 1980; *Opornye*..., 1983; *Mezhdunarodnyi*..., 1991; *Pan-Arctic*..., 2000). Outcrop 6a is Permian rocks studied in this paper.

Permian sections of the RP (Iosifidi and Khramov, 2002; Shatsillo et al., 2006; Iosifidi et al., 2010) demonstrated a reasonable agreement with the data obtained in the 1960s–1970s. This is due to the fact that many red rocks have a small viscous component and single-component NRM composition. In this paper, the new results of studying the Permian rocks of the Subpolar Urals (the Kozhim River) and RP (the Kama River region) are presented and the maximal total cumulative amplitudes of the thrusts are estimated for the structures of the Kozhim River.

OBJECTS AND METHODS OF STUDY

Subpolar Urals. The study objects are samples of the collections of 1993 and 2012 taken from the Paleozoic sequences in the basin of the Kozhim River (Fig. 1). The Asselian deposits are sampled from the skeletal hill (Fig. 1b, outcrop 10) and represented by massive bioclastic and bioherm limestones (Antoshkina et al., 2000). Ten rock pieces are gathered in the middle part of the hill along the coast of the Kozhim River with an interval of 10–15 m. The collection of the Kungurian (Filippovskii and Iren'skii horizons) and Ufimian rocks (Solikamskii horizon) is taken in the outcrops of the left bank of the Kozhim River 2 km downstream from the railroad bridge across the river (outcrop 6a). In this work, we studied siltstones, mudstones, and sandstones sampled from the upper part of the Filippovskii and lower part of the Iren'skii horizons (34 rock pieces, 78 m of the section sampled), the upper part of the Iren'skii horizon (25 rock pieces, 47 m of the section sampled), and the upper part of the Solikamskii horizon (19 rock pieces, 10 m of the section sampled). The bedding elements of the stratum vary within 309°–315° in the dip azimuth and 78°–83° in the dip angle; the average coordinates of sampling are 65.75° N, 59.55° E.

Russian Platform. We reinvestigated the Middle Permian rocks of Kazanian and Ufimian age sampled in the coastal outcrops of the Kama River (collection of A.N. Khramov, 1956) and Ufimian rocks sampled in Perm, on the left bank upstream of the Perm I railroad station and in the region of the Monastyrka village, opposite to the town of Osa. The rocks selected for the study are brown-red clays (12 samples, total thickness 50 m). The Kazanian rocks are sampled on the right bank of the Kama River 3 km downstream of the Tabory village (Khramov, 1958; 1963). Here, the samples also represent red-brown clays with interbeds of sandstones and marls (20 rock pieces, total thickness of the section is 60 m). The rock bedding is subhorizontal and the average sampling coordinates are 57.7° N, 55.6° E.

The laboratory paleomagnetic studies and processing of the obtained results followed the standard procedure (*Paleomagnitologiya*, 1982; Khramov, 1987). Thermal demagnetization of the rocks was carried out

Table 1. Limits of variations in NRM and K_m of studied rocks. For comparison, data for Artinskian–Sakmarian deposits of Kozhim River (Iosifidi and Khramov, 2013) are presented; line 1, Kama River; lines 2–5, Kozhim River

No.	Age of rock	Number of measured samples	Range of NRM variations, mA/m	Average NRM, mA/m	Number of measured samples	Range of variations in K_m, 10^{-6} SI units	Average K_m, 10^{-6} SI units
1	$P_{1\text{-}2}$ u, kz	53	0.3–12.2	4.6	25	13–140	63
2	P_1a	14	0.1–0.8	0.4			
3	P_1u	39	3.8–36.0	16.0	43	402–12479	4352
4	P_1k	120	0.8–46.0	9.2	109	291–11843	1733
5	P_1s-ar	71	0.1–4.8	0.7	59	44–279	127

with the use of the equipment placed into five-layer permalloy shields (original design by VNIGRI) and thermal demagnetizer TD48 (USA). NRM was measured by JR-4 and JR-5 magnetometers (AGICO, Czech Republic). Magnetic susceptibility measurements were conducted by the KLY-2 and KLY-3 kappa-bridges (AGICO, Czech Republic). For analyzing the obtained data and visualizing the results in graphical form, we used the program packages of (Popov, 2007; 2008; Enkin, 1994, Chadima and Jelinek, 2009). The fold test was used in the modification (Watson and Enkin, 1993). In data processing, the statistics were calculated on the level of the rock pieces (stratigraphic levels).

EXPERIMENTAL DATA

Magnetic properties. The values of the scalar magnetic characteristics (natural remanent magnetization (NRM) and magnetic susceptibility (K_m)) for the rocks of the Kozhim River are presented in Table 1. The paleomagnetic studies in the northeastern part of the Pechora Plate are complicated by the fact that most rocks here are abundant with plant remains and coal-bearing interbeds and contain iron sulfides. Their disintegration in the course of thermal demagnetization typically results in the crystallization of new magnetic minerals. In some cases, these peculiarities of the rocks preclude the complete isolation of the characteristic NRM component during thermal demagnetization (Iosifidi and Khramov, 2013). Similar problems also arose during the thermal demagnetization of the rocks from the new Kozhim River collection. The occurring transformations are illustrated by the examples of K_m behavior during heating and cooling (Fig. 2). All the samples experience strong chemical transformations after heating to 300–350°C. This is clearly seen in the behavior of the K_m curves during repeated heatings and coolings (the first heating to 350°C, second heating to 510°C, and third heating to 610°C) for sample K137. Nevertheless, the characteristic NRM component can still be isolated in most samples. The deblocking temperatures of the NRM components for the rock samples in which it was possible to isolate the characteristic component are 350–570°C. This suggests that the characteristic NRM component is likely to be largely carried by the magnetic minerals of the titanomagnetite series. In the rocks from the outcrops on the Kama River, strong chemical transformations are not observed (Fig. 2, sample 3234). The NRM deblocking temperatures in the samples where it was possible to isolate the characteristic magnetization component are 400–670°C. The main carrier of the characteristic NRM component is hematite.

Component analysis of NRM. The isolation of the NRM components was conducted by the stepwise demagnetization procedure. As seen from the Zijderveld diagrams (Fig. 3), all the samples have two NRM components. The NRM of all the samples contains the postfolding (or close to postfolding) component A. In the rocks from the Kozhim River outcrops, component A is isolated in the low-temperature interval (100–350°C) and is most likely associated with the viscous magnetization of the rocks in the present or Cenozoic geomagnetic field. The distribution of component A for all the studied samples is shown in Fig. 4a and its statistical characteristics are presented in Table 2. The fold test for the entire distribution is indeterminate (Table 2). The high-temperature component B is isolated in the temperature intervals 350–480 and 350–570°C. The distribution of components B for the Permian rocks from the Kozhim River outcrops is shown in Figs. 4b–4d and their statistical characteristics are presented in Table 2. For the Asselian rocks, the conversion to the stratigraphic coordinate system is carried out based on the average bedding elements of Late Carboniferous deposits from outcrop 7, Fig. 1, where the Carboniferous layers are overlain with unconformity by the Permian units of the Asselian age which are located above the skeleton hill (Antoshkina et al., 2000).

In the samples from the outcrops on the Kama River, component A is poorly identified and has a chaotic distribution which can be accounted for by the long storage of the samples under the laboratory conditions and low magnetic viscosity of these rocks. The high-temperature component B is isolated in the temperature intervals 400–650 and 500–670°C. The dis-

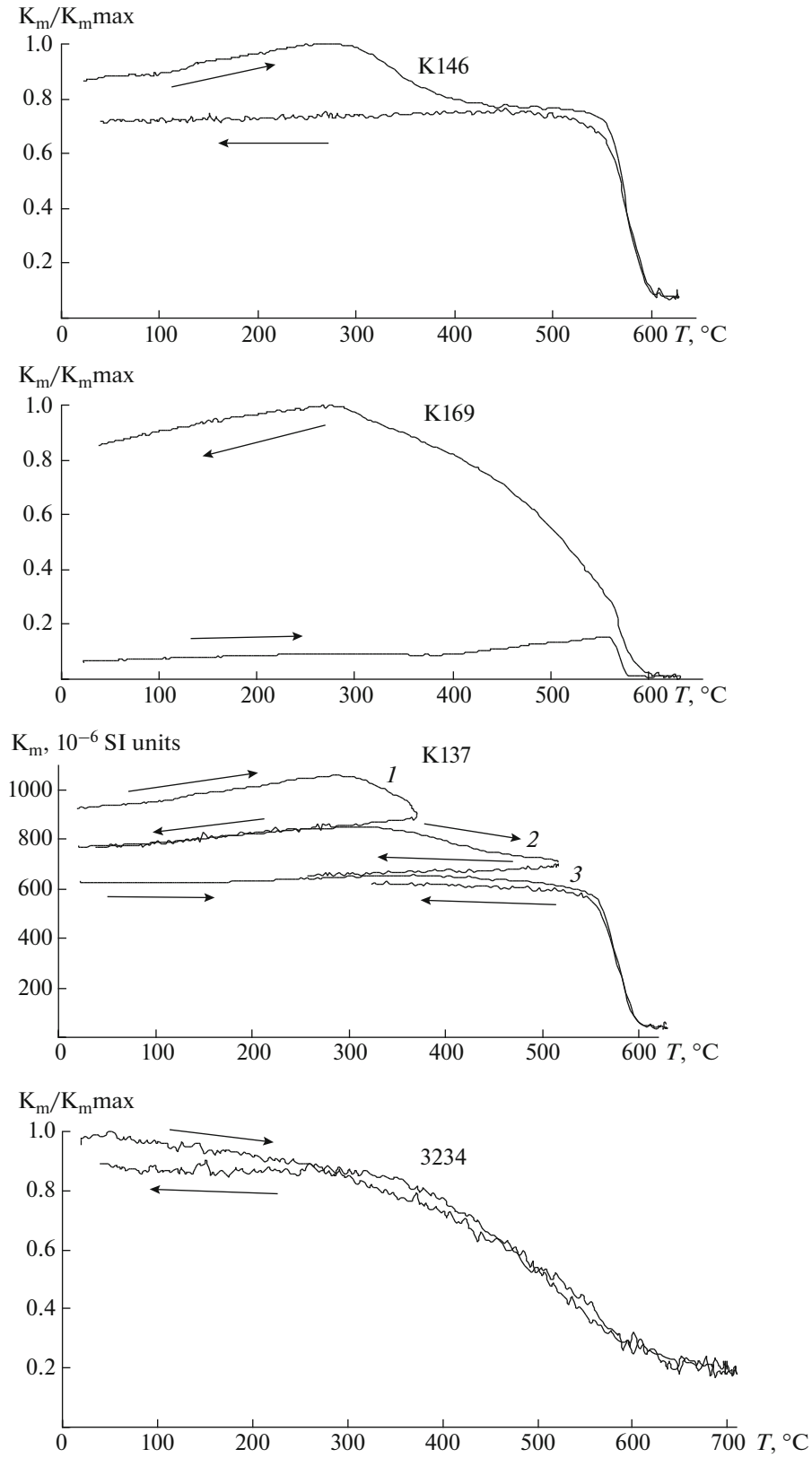


Fig. 2. Examples of thermomagnetic analysis of Permian rocks: curves of variations in magnetic susceptibility under heating (right arrow) and cooling (left arrow). Samples K146, K169, K137 from Kozhim River; sample 3234 from Kama River.

Table 2. Paleomagnetic directions and poles for Permian deposits of Kozhim and Kama rivers

No.	Age of rock	CS	*N*/*n*	D°	I°	*K*	α_{95}°	Φ,° N	Λ,° E	dp°	dm°	φ_m°	Test
	Component A (Kozhim River, 65.76° N, 59.6° E)												
1	P_{12}a-k-u	g	83/154	331	65	12	5	66	293	7	8	47	F~, 0.4 ± 0.3
2		s		321	−10	8	6						
	Component B (Kozhim River, 65.76° N, 59.6° E)												
3		g	9/17	289	−37	56	7						
4	P_1a	s		265	−55	56	7	34	138	7	8	36 ± 4	
5	P_1k	g	39/64	291	1	38	4						
6		s		250	−64	52	3	49	143	4	5	46 ± 2	F+, 0.8 ± 0.2
7	P_2u	g	18/28	287	7	70	4						
8		s		251	−64	81	4	48	142	5	6	46 ± 3	F+, 0.9 ± 0.8
9	5 + 7	g	57/92	290	3	41	3						
10		s		250	−64	60	3	49	143	4	5	46 ± 2	F+, 0.8 ± 0.2
11	Syn 87%	s		265	−58	61	3	37	136	3	4	39 ± 2	
	Component B (Kama River, 57.7° N, 55.6° E)												
12	P_1u		10/17	218	−48	42	8						
13	P_2kz		18/34	231	−48	91	4						
14	P_{1-2}		28/51	226	−48	56	4	47	168	3	5	29 ± 5	

CS is coordinate system (g, geographic; s, stratigraphic); *N*, number of hand samples; *n*, number of specimens; D°, I°, declination and inclination of average directions of components; *K*, concentration parameter of vectors; α_{95}°, confidence circle radius with 95% probability for average direction; Φ°, Λ°, latitude and longitude of paleomagnetic pole; dp°, dm°, half-axes of confidence oval; φ_m°, paleomagnetic latitude; F+(−), positive (negative) fold test after (Watson and Enkin, 1993); hand sample level statistics.

tribution of component B for the Permian rocks from the Kama River outcrops is shown in Figs. 4e–4f and the statistical characteristics are indicated in Table 2.

In the previous studies, it was established that the Permian structures of the Subpolar Urals are typically rotated clockwise in the horizontal plane relative to the RP (Iosifidi and Khramov, 2002; 2005; 2013). For comparing the obtained results with the data for the RP, considering the fact that the time frame of the Late Paleozoic activation in the Urals refers to the Early/Middle Permian time boundary, we calculated the average paleomagnetic poles of the RP in the intervals 299–271 and 270—260 Ma (Table 3). According to these data, the RP (the paleomagnetic direction determination for the center of RP is 57° N, 35° E) has rotated clockwise by (9 ± 3)° and moved northwards by (4 ± 3)° over the considered time interval. For comparing the obtained results with the data for the Kazakhstan blocks (KBs), we calculated the average paleomagnetic poles in the time interval of the Early to Middle Permian (Table 4). The KBs (the determination of the paleomagnetic direction is for the KB center with the coordinates 50° N, 70° E) have rotated clockwise by (3 ± 6)° and moved southwards by (1 ± 7)° (the paleolatitude of the point has changed from 29° to 28° N), which is far below the error of paleolatitude determination.

For the Siberian Platform (SP), there are only scarce data in the interval of Early to Middle Permian; therefore, we used the determinations in the P_1–T_1 interval (Table 5). The determination of the paleomagnetic pole for the SP center (the point with the coordinates 65° N, 105° E) obtained by recalculating the average paleomagnetic pole for this time interval gives the paleolatitude of 65° N for the Siberian Platform.

For the characteristic and synfolding NRM components of the studied structures of the Kozhim River, was calculated the angles of the horizontal rotation relative to the RP (Table 6). The data for the Permian (P_1, Artinskian, and Sakmarian stages) and pre-Permian rocks (S, D, and C_1) are borrowed from (Iosifidi and Khramov, 2013). Under the rotation relative to the Eulerian pole, which is used in this work for the Subpolar Urals (Φ = 68° N, Λ = 58° E), the total amplitude of the displacement of the Permian structures due to the thrusting processes (rotation angle is 35°) is 160 ± 15 km. For the pre-Permian structures (rotation angle is 39°), the displacement amplitude is 180 ± 15 km. An interesting observation is the change in the relative positions of the studied structures. The distance between the midpoints of the studied Permian structures is 7 km, whereas after the reconstruction this distance is 18 km. The distance between the pre-Permian (S, D, and C_1) and Permian structures the

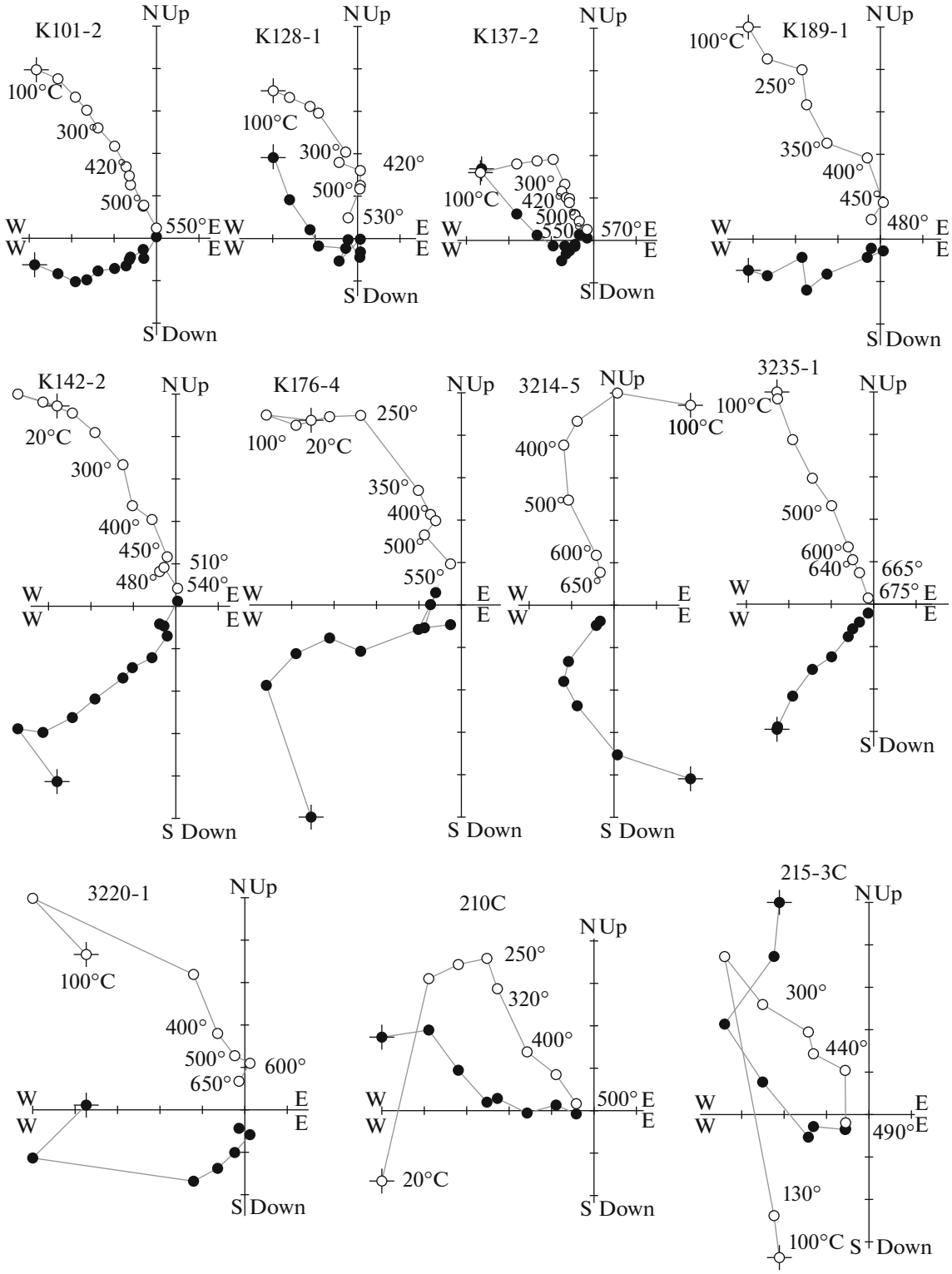


Fig. 3. Examples of stepwise thermal; demagnetization of Permian samples from Kozhim River: samples 210C, 215-3C of Asselian rocks; samples K101-2, K128-1, K137-2, K142-2 of Kungurian rocks; samples K176-4 and K189-1 of Ufimian rocks; Kama River: samples 3214-5, 3220-1 of Ufimian rocks and sample 3235-1 of Kazanian sediments. Zijderveld diagrams are presented in stratigraphic coordinate system; open and filled circles are vector projections on horizontal and vertical plane, respectively. Zijderveld diagrams are constructed with use of Remasoft 3.0 program (Chadima and Hrounda, 2006).

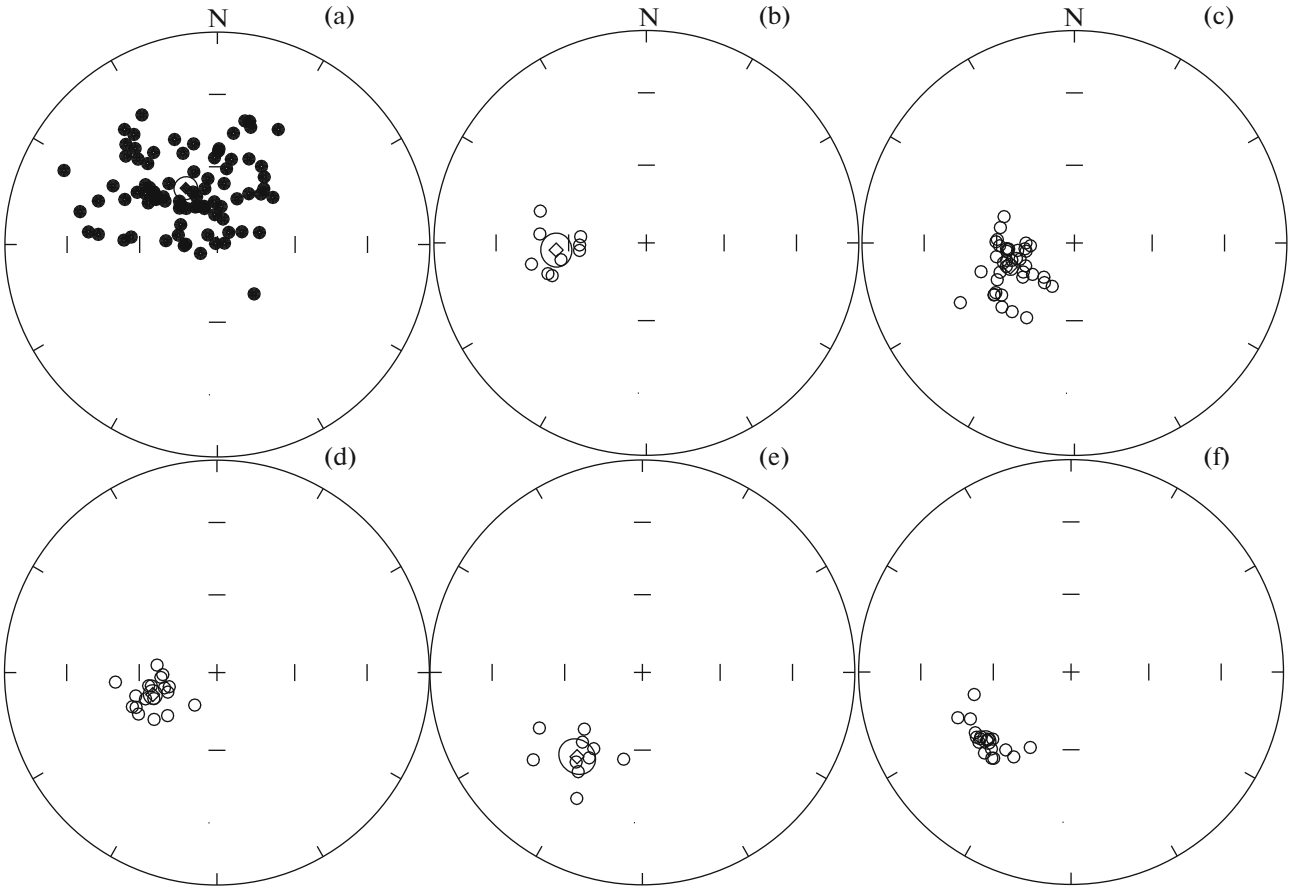


Fig. 4. Distribution of NRM components after demagnetization: (a) component A (geographical coordinate system); component B (stratigraphic coordinate system); (b) Asselian; (c) Kungurian; (d) Ufimian sediments of Kozhim River; (e) Ufimian, (f) Kazanian deposits of Kama River; empty circles are vector projections on upper hemisphere, filled circles are vector projections on lower hemisphere.

corresponding distances are 9 and 22 km, respectively. Based on the data for the Russian and Siberian platforms and the KBs (Tables 3 to 5), we reconstructed the positions of these structures for the Early and Middle Permian (Fig. 5).

Anisotropy of magnetic susceptibility. In the Ordovician, Silurian, Devonian, and Early Permian Artinskian rocks of the Kozhim River, the magnetic susceptibility (K_m) and the degree of its anisotropy (P) have the lowest values (Table 7). In these rocks, the distributions of the directions of the principal axes of the anisotropy ellipsoid of magnetic susceptibility are characteristic of the quiet sedimentation conditions (Fig. 6). The fairly broad scatter in the directions of the principal axes of the ellipsoids is due to the large measurement errors of these parameters against the very low values of magnetic susceptibility.

In the Kungurian and Ufimian rocks, during the field works, we revealed the intervals where the accumulation of sediments occurred in different hydrodynamic conditions (Table 7, Fig. 7). These conditions could be reflected/are actually reflected in the pattern of the directional distribution of the principal axes of magnetic susceptibility of anisotropy ellipsoids.

A significant deviation of the distributions of the semiaxes of anisotropy of ellipsoids from the normal distribution for sediments of these stratigraphic stages can be caused by the following factors:

■ The sedimentation conditions guiding the formation of a certain texture of the rock and, consequently, the corresponding directions of the major, intermediate, and minor axes K1, K2, and K3 of magnetic susceptibility of an anisotropy ellipsoid;

■ The significant secondary impacts on the rock (Zhuravlev et al., 2013). One of these impacts is pressure during the folding processes, which can lead to the deformation of the anisotropy of an ellipsoid and change the directions of its axes. The different magnitude of the pressure can change the K1 direction into coaxial and orthogonal to the direction of motion (Igoshin, 1986; Zhuravlev et al., 2013). The pressure can also significantly change the NRM direction of the rocks (Igoshin, 1986; Zhuravlev et al., 2013).

Table 3. Paleomagnetic poles of RP in Early to Middle Permian

No.	φ, ° N	λ, ° E	Φ, ° N	Λ, ° E	dp°	dm°	Age, Ma	RESULTNO/REFNO in GPMDB
1	55.0	52.0	55	173	3	5	260	4805/2016
2	48.0	38.0	55	171	10	17	260	4880/2029
3	55.5	50.5	56	165	4	5	260	4694/2034
4	56.0	52.0	50	169	5	7	260	480182015
5	54.5	52.5	50	169	6	10	261	4807/2016
6	58.1	57.4	50	156	4	6	261	4889/2015
7	56.0	52.0	47	166	3	5	261	4800/2015
8	55.0	53.0	45	169	5	8	261	4808/2015
9	55.0	53.0	45	164	3	5	261	4809/2015
10	54.0	52.0	45	163	5	8	261	4810/2087
11	56.0	52.0	46	175	1	2	263	4798/2015
12	55.0	53.0	45	170	2	4	264	4803/2015
13	55.0	53.0	46	170	3	5	264	4804/2015
14	47.5	38.0	40	180	5	10	265	6754/2033
15	56.0	51.0	43	168	4	7	265	4693/2622
16	56.0	51.0	44	169	2	4	266	4799/2015
17	57.7	55.6	47	168	3	5	268	/1
18	56.0	53.0	43	167	2	3	270	4816/2042
19	56.0	52.0	46	165	3	5	270	4817/2015
20	59.0	57.0	40	170	5	9	270	4712/2015
21	55.0	53.0	45	180	4	6	270	4818/2015
22	58.0	57.0	40	162	3	4	270	4812/2182
23	P_2	$N = 22$	47	169		$K = 186$	$A_{95} = 2°$	$D = 215°$, $I = -40°$, $\varphi_m = (23 \pm 3)°$
24	58.5	56.5	37	160	13	22	271	4744/2042
25	59.0	57.0	34	162	8	14	271	4713/2042
26	58.5	56.5	37	160	13	22	272	4814/2016
27	54.8	56.7	44	160	3	4	272	/2
28	57.6	57.3	49	162	2	3	274	4891/2087
29	57.5	57.5	41	151	3	4	274	4892/2087
30	57.5	57.5	45	158	3	4	274	4890/2087
31	48.0	38.0	43	169	2	4	293	4685/877
32	49.0	38.5	36	163	2	4	293	4680/877
33	49.0	38.0	35	163	5	10	294	4679/877
34	48.0	38.0	41	167	6	12	294	4684/877
35	48.0	38.0	37	164	3	5	295	4686/877
36	48.5	38.0	40	167	2	4	296	4681/877
37	48.5	38.5	38	155	1	2	296	4682/877
38	49.0	38.5	36	159	4	8	297	4676/5501
39	48.5	38.7	43	164	2	3	297	8937/2201
40	48.5	38.0	39	159	5	10	297	4677/3519
41	48.0	38.0	39	161	3	5	298	4688/877
42	P_1	$N = 18$	40	161		$K = 248$	$A_{95} = 2°$	$D = 224°$, $I = -35°$, $\varphi_m = (19 \pm 3)°$

References: φ, °N; λ, °E, coordinates of sampling; A°_{95}, confidence circle radius at 95% probability for direction of pole; indices of determinations of Global Paleomagnetic Database (GPMDB) (McElhinny and Lock, 1996; Pisarevsky, 2005); References: 1. this work; 2. (Musin, 1991); paleolatitudes are calculated onto coordinates 58° N, 39° E. Remaining designations are described in Notes to Table 2.

Table 4. Paleomagnetic poles of KBs in Early to Middle Permian

No.	φ, ° N	λ, ° E	Φ, ° N	Λ, ° E	dp°	dm°	Age, Ma	RESULTNO/ REFNO in GPMDB
1	51.7	67.8	53	161	3	4	260	/1
2	49.5	70.5	53	187	5	8	260	7635/2933
3	49.5	70.5	44	196	5	8	260	7636/2933
4	47.5	80.7	43	172	4	4	260	8963/3529
5	52.0	67.5	55	166	5	7	265	4843/2070
6	47.5	80.6	43	172	3	4	265	/2
7	P_2	$N = 6$	49	176		$A_{95} = 7°$		
8	47.9	75.3	56	181	4	5	271	/3
9	52.0	67.5	55	166	5	7	271	4843/2070
10	52.0	67.5	51	160	6	9	275	4690/2032
11	48.0	75.6	42	179	5	3	280	/3
12	52.0	67.5	51	160	6	9	280	4690/2032
13	48.6	73.8	47	197	9	16	285	5018/2181
14	48.0	74.0	55	195	5	7	285	5019/2181
15	P_1	$N = 7$	52	177		$A_{95} = 7°$		

References: 1. (Bazenov et al., 2008); 2. (Levashova et al., 2003); 3. (Abrajevitch et al., 2008); remaining designations are described in Notes to Table 2.

Table 5. Paleomagnetic poles of Siberian Platform in P_1–T_1 interval

No.	φ, ° N	λ, ° E	Φ, ° N	Λ, ° E	dp°	dm°	Age	RESULTNO/ REFNO in GPMDB
1	66.1	111.6	51	158	9	10	$P–T_1$	4578/1997
2	64.1	114.7	52	144	14	14	$P–T_1$	4572/1997
3	63.7	115.4	58	137	14	14	$P–T_1$	4573/1997
4	64.5	130.0	47	135	13	14	$P_{2–3}$	710/1
5	69.0	88.0	59	149	7	7	$P_{2–3}$	4718/1990
6	71.5	116.0	51	179	9	11	P–T	4839/2067
7	70.4	120.7	39	143	13	15	P–T	4579/1997
8	63.0	112.0	52	140	6	6	P–T	4524/1986
9	65.2	111.2	39	139	13	14	$P_{1–2}$	4574/1997
10	58.5	102.5	56	122	12	12	P_1	4695/880
11	P_1–T_1	$N = 10$	51	145		$A_{95} = 6°$		

References: 1. (*Paleomagnitnye*..., 1971); remaining designations are described in Notes to Table 3.

Therefore, the interpretation of the measurements of the anisotropy of magnetic susceptibility for the Early Permian parts of the Kungurian and Ufimian sections is ambiguous. On one hand, declinations of K1 frequently coincide with the trends of the fold hinge lines (Figs. 7a–7d). This is probably at least partly associated with pressure. However, on the other hand, this distribution of K1 and the simultaneous elongated distribution of K3 are characteristic of the specific sedimentation conditions, e.g., in the zone of deltaic flows (Fig. 7a) and lagoons (Fig. 7b), under the action of waves (Fig. 7c). One can easily imagine the formation scenario of this anisotropy: the deltaic flows are subperpendicular to the coastline; tidal and ebb impacts have the same direction (including in the lagoons), and plate collisions and folding also typically occur in this direction. All these nonsimultaneous processes can align the major K1 axis of the magnetic

Table 6. Calculation of horizontal rotation angles of River Kozhim structures relative to RP

No.	Age of rock	Sampling coordinates, ° N , ° E	*N*	*D*°	*I*°	*K*	α_{95}°	φ_m°	D_C°	φ_{mc}°	$\Delta_{-\varphi m}^{\circ}(\varphi_{mc}^{\circ} - \varphi_m^{\circ})$	Δ°	Amplitude, km
	Characteristic NRM components (Kozhim River, 65.76° N, 59.6° E)												
1	P_1a	65.70, 59.70	9	265	−55	56	7	36 ± 4	232	36	0	−33	
2	P_1s-ar	65.71, 59.68	28	267	−53	40	4	34 ± 7	232	36	2 ± 3	−35 ± 4	
3	Pk-u	65.76, 59.6	57	250	−64	60	3	46 ± 2	232	36	−10 ± 3	−18 ± 3	
4	$P_1(3 + 4)$	65.7, 59.6	85	257	−60	39	3	41 ± 6	232	36	−5 ± 3	−25 ± 3	
5	S,D,C_1	65.67, 59.71	100	256	−59	47	2	40 ± 3	232	36	−4 ± 2	−24 ± 3	
	Synfolding NRM components (Kozhim River , 65.76° N, 59.6° E)												
6	P_1s-ar, 90%	65.71, 59.68	28	268	−55	44	4	36 ± 2	232	36	0	−36 ± 3	165 ± 15
7	Pk-u, 87%	65.76, 59.6	57	265	−58	61	3	39 ± 2	232	36	−3 ± 3	−33 ± 4	150 ± 20
8	$P_1(6 + 7)$, 85%	65.74, 59.64	85	267	−56	54	2	37 ± 3	232	36	−1 ± 2	−35 ± 3	160 ± 15
9	S,D,C_1, 76%	65.67, 59.71	100	271	−56	49	2	37 ± 3	232	36	−2 ± 2	−39 ± 3	180 ± 15

References: $D_c^{\circ}, \varphi_{mc}^{\circ}$, deviation and paleolatitude obtained by recalculating of values corresponding to reference pole of RP for Middle Permian (Φ = 47° N, Λ = 169° E, A_{95} = 2°; D = 232, I = −55, K = 222, a_{95} = 2°, φ_m = 36 ± 2) to average point of sampling (65.7° N, 59.6° E); $\Delta_{-\varphi m}^{\circ}$ ($\varphi_{mc}^{\circ} - \varphi_m^{\circ}$), difference in paleolatitudes; Δ°, total rotation angle due to thrusting. Remaining designations are described in Notes to Table 2.

Table 7. Key parameters of studied anisotropy of magnetic susceptibility of rocks. *N*, number of samples; $K_m$min/$K_m$max, minimal/maximal magnetic susceptibility; Pmin/Pmax, minimal/maximal degree of anisotropy of magnetic susceptibility

No.	Age of rock	*N*	$K_m$min, SI units × 10^{-6}	$K_m$max, SI units × 10^{-6}	Pmin, %	Pmax, %
1	O	24	−8	171	1.2	8.7
2	S	13	86	334	1.1	6.7
3	D	18	−1	83	0.7	11.6
4	P_1ar	25	50	223	0.4	5.8
5	P_1k–P_1u – wave field	94	440	12440	0.9	13.5
6	P_1k–P_1u – deltaic channels	35	470	11840	1.0	13.7
7	P_1k–P_1u – lagoon	15	397	1295	1.6	7.4
8	P_1k–P_1u – concretions	6	292	782	2.3	8.9

particles parallel to the coastline, which is what we actually observe. At the same time, if the K1 were solely related to pressure alone, the minor axis of the ellipsoid would be oriented, as a result of compression, everywhere identically, in the direction perpendicular to pressure. However, this is not actually observed in the age interval of the Kungurian and Ufimian rocks: this axis is distributed along the great arc circle within an aperture of up to 180 degrees (Figs. 7a–7d).

In this case, the results are likely to suggest the preservation of the primary texture of the rocks and, if so, the strike of the ancient coastline in the NE–SW direction in the present coordinates. It is also worth noting that if the K1 directions in the Permian rocks of the Kungurian and Ufimian age (Fig. 7) were only connected with pressure, this influence should have also manifested itself in the more ancient Ordovician–Early Permian Artinskian rocks, which is not the case (Fig. 6). The absence of a correlation between the directions of the NRM and principal axes of the ellipsoids of the anisotropy of magnetic susceptibility is also understandable. The NRM directions were mainly isolated in the high-temperature interval of the magnetic treatment, whereas the anisotropy of the magnetic susceptibility is measured in the alternating magnetic field in the interval of low coercivity. Hence, magnetization and anisotropy are carried by the particles with different sizes and their spatial orientation is

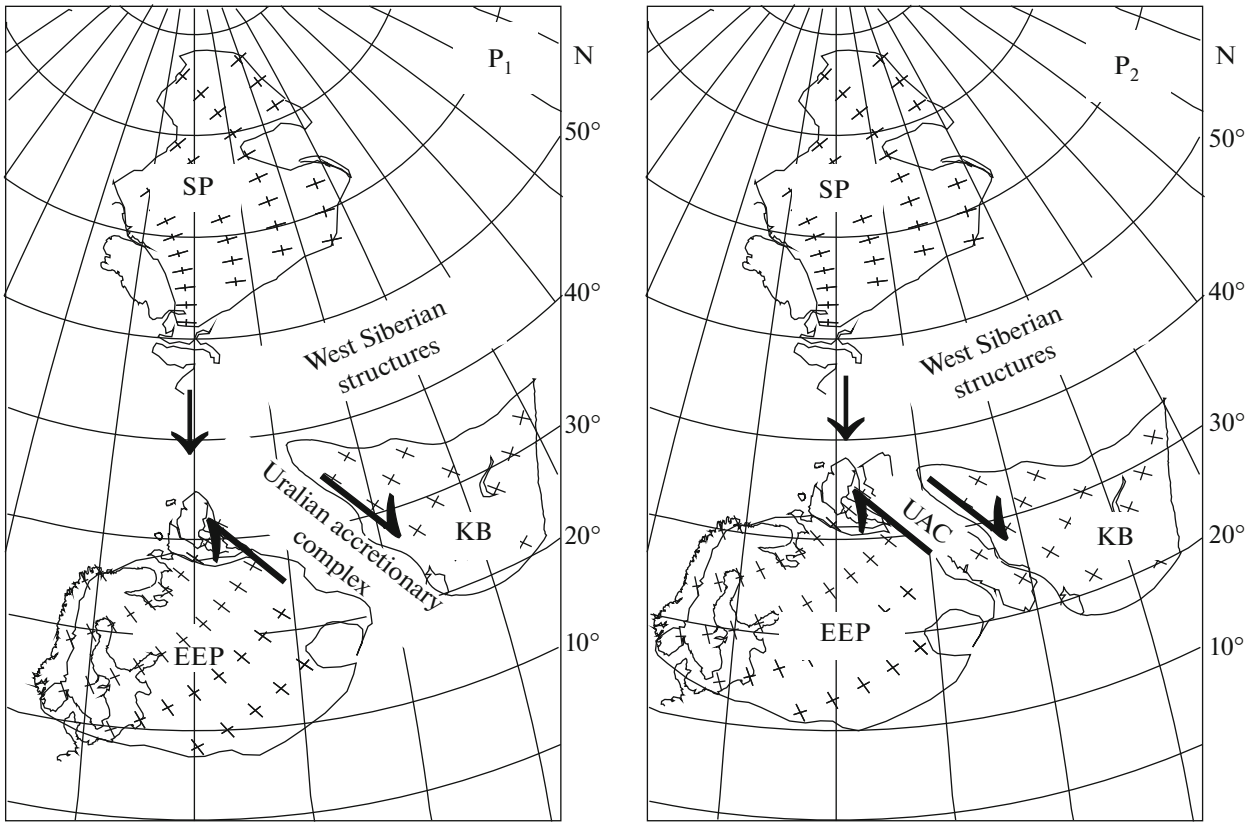


Fig. 5. Relative position of tectonic plates and blocks of North Eurasia in Early to Middle Permian. SP, Siberian Platform; EEP, East European Platform; KB, Kazakhstan blocks; UAC, Uralian accretionary complex, UAC position in reconstruction for Middle Permian is calculated from paleomagnetic pole for South Urals $\Phi = 46°$, $\Lambda = 170°$, $A_{95} = 4°$ (Bazhenov et al., 2008). Reconstruction or figure is constructed by GMAP program (Torsvik et al., 1990).

in the first case associated with the influence and direction of the magnetic field and in the second case with gravitation and hydrodynamic impacts.

DISCUSSION

According to the existing paleomagnetic data for the RP, in the interval of 260–299 Ma the paleolatitude has changed by 4 degrees (Table 3). The average paleomagnetic pole for this time interval has the coordinates $N = 40$, $\Phi = 44°$ N, $\Lambda = 165°$ E, and $A_{95} = 2°$. The data selection of the paleomagnetic poles of the EEP of (Torsvik et al., 2012) in the interval 260–299 Ma gives the similar position of the average pole: $N = 60$, $\Phi = 44°$ N, $\Lambda = 165°$ E, and $A_{95} = 2°$. At the same time, the average poles for the Early and Middle Permian are different. In (Torsvik et al., 2012), the authors mainly use the data beyond the RP (only four determinations for the RP are considered). The paleomagnetic poles for P_1 and P_2 of the EEP (Torsvik et al., 2012) have the following positions: $N = 51$, $\Phi = 43°$ N, $\Lambda = 167°$ E, $A_{95} = 2°$ and $N = 9$, $\Phi = 50°$ N, $\Lambda = 155°$ E, $A_{95} = 4°$, respectively (for our estimates, we used the original paleomagnetic determinations from the set presented in (Torsvik et al., 2012) without the correction for inclination shallowing), which disagrees with our estimate for the RP (Table 3). On one hand, this is due to the fact that, as mentioned in the Introduction, the data for the RP do not always satisfy the present-day criteria of reliability assessment. On the other hand, in the discussed set of the poles of the RP, 75% of the data for the Middle Permian are obtained by thermal demagnetization up to 500°C (demagnetization code 2–3), whereas the remaining data correspond to the demagnetization by the alternating magnetic field. The difference between the average paleomagnetic poles for Baltica and the RP is maximal for the Middle Permian. In our opinion, the determinations based on the data beyond the RP do not always take into account the local and regional tectonics, whereas in the case of the site sampling, the data can be insufficiently averaged over time. In the further processing of the paleomagnetic data, many authors introduce corrections for inclination shallowing by arbitrarily increasing (almost doubling) the inclination values for the sedimentary rocks. The use of the paleomagnetic poles determined from the Siberian traps for constructing the apparent polar wonder path for the EEP is, in our opinion, also incorrect (the situation replicates the one where the data for Carbonif-

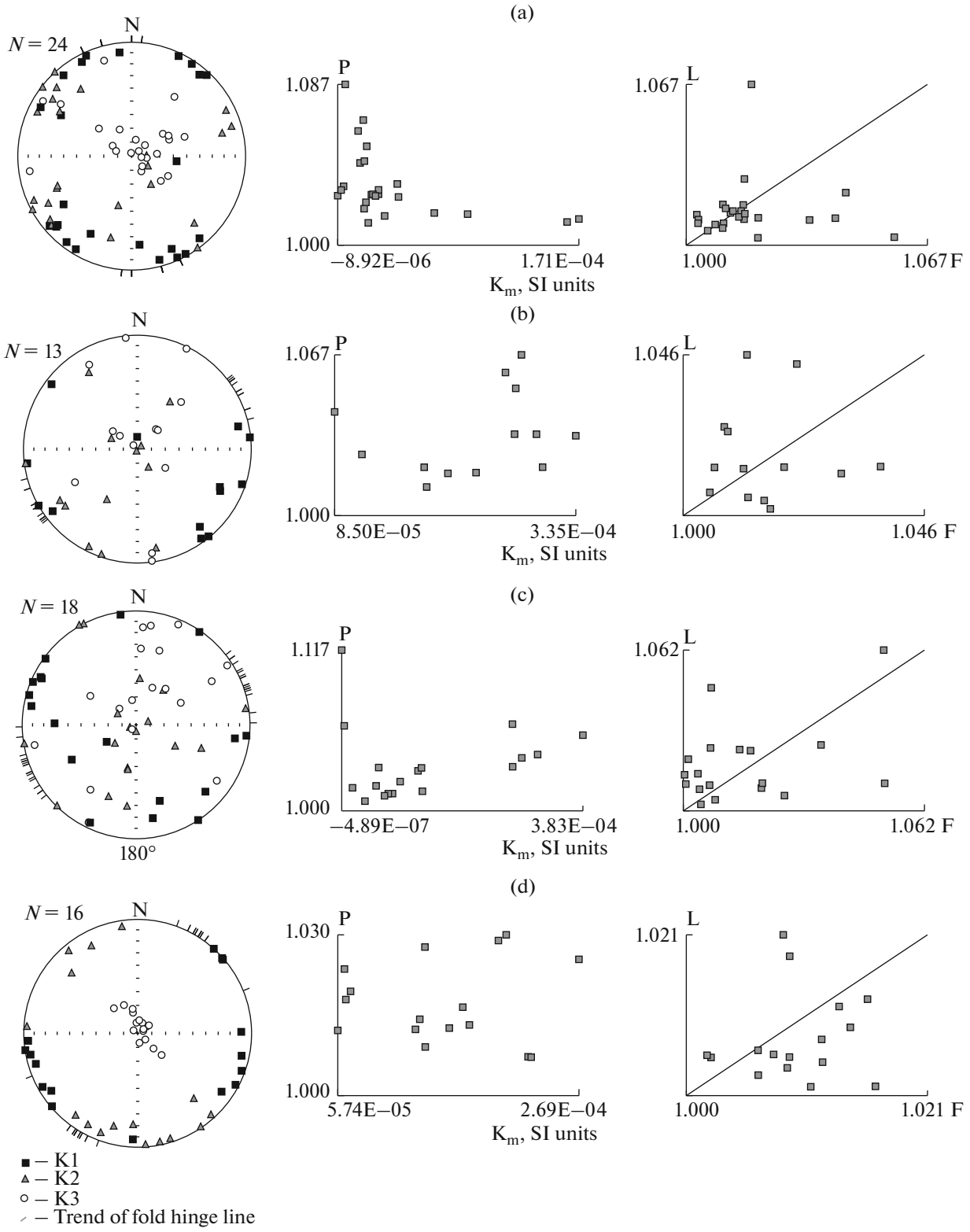


Fig. 6. Distribution of directions of principal axes of magnetic susceptibility ellipsoids: (a) Ordovician; (b) Silurian; (c) Devonian; (d) Early Permian. K1, K2, and K3 are maximal, intermediate, and minimal half-axes of ellipsoid of magnetic susceptibility anisotropy. P is degree of anisotropy; K_m is magnetic susceptibility; F is the oblateness, and L is elongation of the ellipsoid.

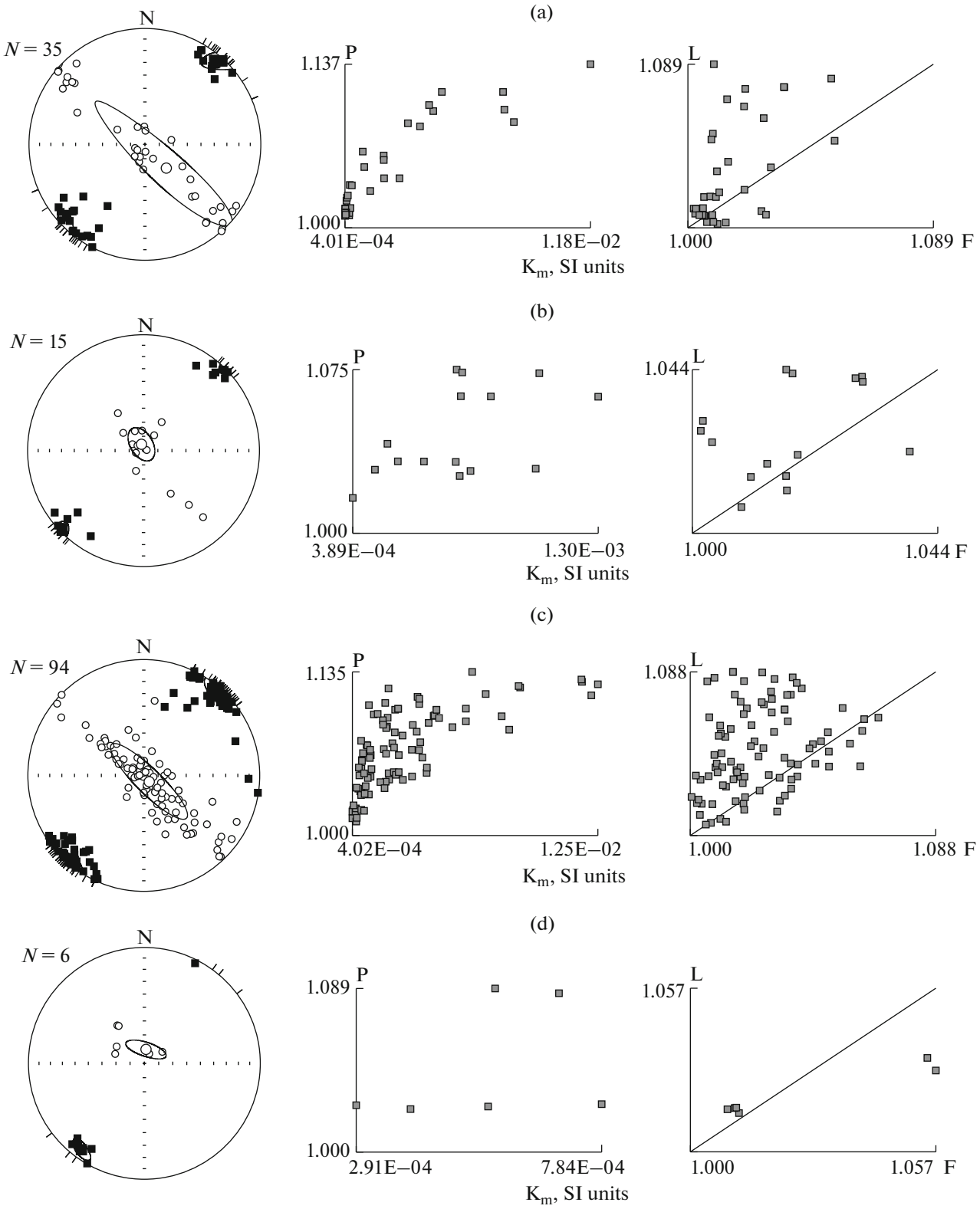


Fig. 7. Distributions of directions of principal axes of magnetic susceptibility ellipsoids in Permian rocks of Kungurian and Ufimian age which have different formation conditions: (a) deltaic channels; (b) lagoonal conditions; (c) conditions of influence of wave field; (d) concretions. Remaining designations are described in Fig. 6.

erous rocks from Spitsbergen were used for the EEP). All this, to some extent, could lead to the dissimilarity in the positions of the average paleomagnetic poles P_1 and P_2 for the EEP (Torsvik et al., 2012) and the RP (Table 3).

The paleomagnetic determination based on the rock material from the Kama River is closely consistent with the paleomagnetic pole for the Middle Permian for the RP (Tables 2 and 3). We note that in (Khramov, 1958; 1963) where the specimens were only subjected to time demagnetization (placed in the magnetic vacuum for a certain time), similar directions were obtained: $D = 220°–230°$, $I = -37°–42°$. The paleomagnetic determinations obtained for the Permian rocks (Kungurian and Ufimian stages) of the Kozhim River section agree with the direction obtained in 2013 for the Early Permian (Artinskian and Sakmarian) rocks of this section (Tables 2 and 3). The characteristic components of the P_1k-u rocks have a higher inclination than for the Artinskian and Sakmarian rocks which reflects the joint northward motion of the RP and Subpolar Urals in the Permian. The significant rotations (by 10°–40°) are associated with the thrusting processes during the EEP collision with the structures of the Urals. A characteristic feature of these processes is that all the structures of the Kozhim River have undergone clockwise rotation. The estimates of the rotation angles of the structures ($\Delta°$) calculated with the use of the average paleomagnetic pole of the RP (47° N, 169° E, $A_{95} = 2°$) for the Middle Permian are presented in Table 6. Here, the values of the rotation angles for most of the objects exceed the determination errors of the directions. The discrepancies are observed in the horizontal rotation angles of the studied structures relative to the RP in the Middle Permian (Table 6). This can be due to the fact that the studied structures pertain to different tectonic plates or are associated with the presence of the synfolding component in NRM. The directions of the characteristic NRM components isolated in the Paleozoic rocks of the Kozhim River sections correspond to the Early to Middle Permian direction of the geomagnetic field. This indicates that the horizontal rotations of the studied structures are most likely to have occurred during the existence of the Kiama hyperchron ($C_2–P_2$). Taking into account the complicated evolution of the region, we may hypothesize the formation of the synfolding component in NRM. The direction of the synfolding components in NRM is identified by the proportional unfolding test. The results of this test are presented in Table 6. The paleomagnetic directions of the synfolding components of NRM become more consistent with each other both in terms of paleolatitude and horizontal rotation angles of the structures. The model of remagnetization of the Paleozoic rocks of the Kozhim River is suggested in (Iosifidi and Khramov, 2013). Inter alia, the synfolding component could appear because, e.g., the pre-Permian rocks were in a tilted position during their Early Permian remagnetization (Eremenko et al., 2009). This hypothesis is also supported by the recalculation of the bedding elements of the structures into the directions of the present and synfolding components. In the synfolding system of the coordinates, the dip angle of the structures decreases compared to the stratigraphic coordinate system and, correspondingly, so does the inclination of the NRM vectors which is in a number of cases clearly overestimated and close to the Triassic NRM directions (Table 3). The question of the probability of Triassic remagnetization of the Paleozoic rocks was discussed in (Eremenko et al., 2009; Iosifidi and Khramov, 2013). For the Permian sediments under the continuing collision of the RP with the Uralian structures, also the fact of molasse deposition on the sloping surface can be accepted. On the left bank of the Kozhim River, opposite to the Monakh Rock, there is an angular unconformity between the Permian rocks of the Asselian age and the Gzhelian limestones.

Considering this, we calculated the total amplitudes of the displacement of the studied structures during the collision between the EEP, the structures of the Urals, Siberian Platform, and KBs from the synfolding NRM components. The amplitudes of the thrusts for the Permian and pre-Permian remagnetized rocks of the Kozhin River are close (Table 6). We note that for the Subpolar Urals, the direction of motion of the thrust fronts has a meridional component in the present coordinates, which counts in favor of the oblique collision of the EEP with the Uralian structures. The process of this oblique collision was accompanied by the development of right-lateral shears which caused the clockwise rotation of the thrust structures relative to the RP. The amplitudes of thrusting of the structures of Subpolar Urals relative to the RP are, on average, 170 km. These processes could probably act in the following chain: during the collision between the EEP, KBs and Siberian Platform, tangential compressions forced the KBs to move southwards. In the course of this motion, the structures of the Cis-Uralian foredeep underwent right-lateral strike-slip displacement relative to the RP, which caused clockwise rotations (Fig. 5). Our estimation of the relative amplitude of the thrusts within the Kos'yu–Rogovskaya depression (across the top of the Carboniferous and Permian sediments) on the geological cross section at the latitude of the Kozhim River (Yudin, 1994, Fig. 29c) gives the value of 40 km. This discrepancy is due to the fact that the geological cross section reflects the last stage of the relative thrusts for the structures of the Kos'yu–Rogovskaya depression, whereas the joint motions of the Uralian structures in the course of their collision with the EEP, KBs, and Siberian Platform are disregarded.

CONCLUSIONS

• The Paleozoic structures of the Near-Urals (the Kozhim River) underwent horizontal clockwise rotations by up to 40° relative to the RP;

• For the structures of the Near-Polar Urals (the Kozhim River), the amplitudes of the thrusts were, on average, 170 ± 15 km as of the Middle Permian;

• The direction of motion of the thrust fronts has a meridional component (in the present coordinates), which testifies in favor of an oblique collision of the Uralian structures with the EEP;

• The probable succession of these processes could be as follows: during the collision of the EEP, KBs, and Siberian Platform, due to the tangential compressions, the KBs moved southwards. Upon this, a right-lateral shear relative to the RP occurred, which caused the clockwise rotation of the structures of the Cis-Uralian foredeep.

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